

01 / RESEARCH OBJECTIVE

FININ: a working reduced prototype

Reproducing news interactions for next-session market prediction

Paper. Wang, Cohen and Ma (2024), *Modeling News Interactions and Influence for Financial Market Prediction*, Findings of EMNLP, pp. 3302–3314 [1].

Research question. Can a model improve a market forecast by first relating headlines to one another, then weighting them using the current market state? FININ combines each headline’s meaning and sentiment, rather than reducing the entire news day to one sentiment score.

Progress assessment: sufficient for a working architectural prototype. The data-to-prediction pipeline runs, all 18 trained checkpoints replay correctly, and the core fusion and attention mechanisms are implemented. The published performance gains and financial conclusions have **not** been reproduced.

Paper architecture versus our implementation

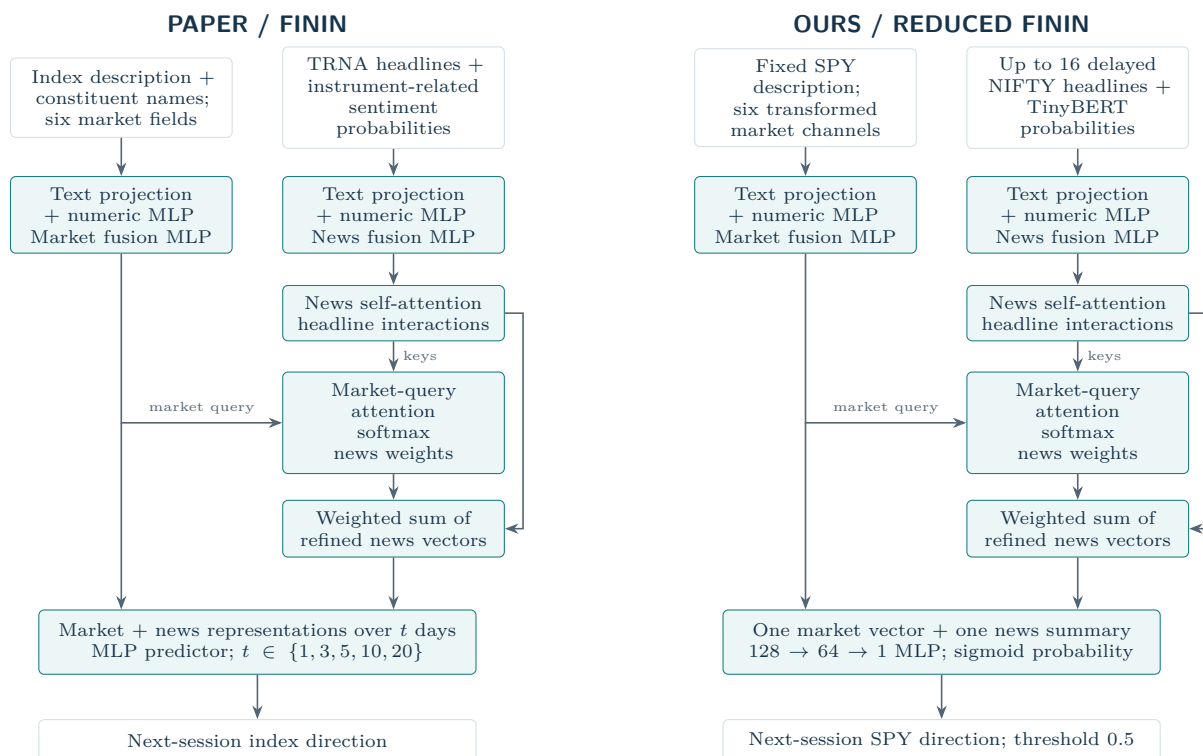


Figure 1. Conceptual redraw of paper Figure 2 and the implemented data flow. Teal blocks retain the paper’s functional structure. Market features also reach the predictor directly; refined news supplies both attention keys and the weighted-sum values.

What is shared. Within each model, market text and headline text use the same frozen language model and learned projection. Market and news numeric/fusion MLPs remain separate. The paper selects RoBERTa for its main comparisons; ours freezes BGE-small and trains 91,905 downstream parameters.

02 / DATA AND FORECAST TIMING

Same task family, different evidence

Dimension	Paper [1]	Our prototype [2; local audit]
Market	S&P 500 and NASDAQ 100 indices	SPY ETF, a proxy for the S&P 500; one market
Period	2003–2018	News dates: Jan. 2010–Sept. 2020; last target: 23 Sept. 2020
News	TRNA: over 2.7 million items; 24–1,953 per day	NIFTY: 436,570 raw headline occurrences over 2,111 dated rows; 33,673 selected slots, 32,286 unique texts
Selection	Aims to use all available daily news	Within-day exact deduplication; deterministic SHA-256 ordering; cap of 16
Sentiment	Positive/neutral/negative scores about mentioned financial instruments	Three text-sentiment probabilities from a frozen finance TinyBERT; a proxy, not Reuters analytics
Market inputs	Open, high, low, close, adjusted close, volume; description and constituent names	Five relative-price channels + $\log(1 + \text{volume})$; training-only standardization; fixed description, no constituent list
Text encoding	Compares RoBERTa, FinBERT, BGE, Llama2; main results use RoBERTa	BGE-small-en-v1.5: 384 dimensions, first-token pooling, no vector normalization; 128-token cap
Evaluation	Ten 500-day windows, stepped by 391 days; chronological 80/10/10 split per window	One chronological source split; four boundary-crossing outcomes removed; three training seeds
Scope	1/3/5/10/20 input days; accuracy, PnL, Sharpe ratio	One input day; accuracy, balanced accuracy, log loss; financial metrics deferred

Counts of raw occurrences, selected slots and unique texts measure different things. NIFTY here denotes the *News-Informed Financial Trend Yield* dataset, not India’s NIFTY index.

Forecast time is explicit

Let n be the news date, d its next trading session, and q the session after d :

$$\underbrace{R_n}_{\text{previous-session news}} + \underbrace{M_d}_{\text{prices through forecast close}} \rightarrow \hat{P}(C_q > C_d), \quad y_d = \mathbf{1}[C_q > C_d].$$



We delay the news because individual publication times are unavailable. This reduces timing risk under correct date buckets; it cannot prove historical availability. The horizon remains **one trading session**, distinct from the input lookback. Source prompts, answers and future prices never enter the feature tensors.

Retained split	Dates	News slots	Up targets	Up share
Training	1,475	23,592	807	54.7%
Validation	315	5,040	172	54.6%
Test	317	5,041	184	58.0%

Coverage shift. Raw median headlines/day fall from 78 to 54 to 26 across the splits. The cap controls computation, but cannot equalize relevance or restore omitted stories. All 2,111 supplied NIFTY returns match the next-session SPY return; targets are nevertheless rebuilt for our delayed-news schedule.

03 / EXPERIMENTAL EVIDENCE

A completed experiment with a negative result

Fixed protocol. Six trained variants, seeds 17/42/73; identical dates, selected headlines and frozen features. Width 64; two affine layers per MLP; one attention head per layer; predictor dropout 0.1. Adam, learning rate 0.001, batch size 16, binary cross-entropy, gradient clipping at 1.0, up to 30 epochs; early stopping after five epochs without sufficient validation-loss improvement. Checkpoints use validation loss; direction threshold is 0.5.

Held-out results: 317 dates

Model / reference	Accuracy %	Balanced acc. %	Log loss
Always up (direction only)	58.04	50.00	—
Training-prior probability [†]	58.04	50.00	0.6824
Prices only	58.25 ± 1.31	50.91 ± 1.50	0.6885 ± 0.0053
Prices + mean sentiment	56.89 ± 0.48	49.91 ± 0.64	0.6884 ± 0.0053
Mean-pooled fused news	58.04 ± 0.00	50.00 ± 0.00	0.6823 ± 0.0007
No news self-attention	58.04 ± 0.00	50.00 ± 0.00	0.6816 ± 0.0008
No sentiment input	58.04 ± 0.00	50.00 ± 0.00	0.6814 ± 0.0004
Reduced FININ	58.36 ± 0.55	50.41 ± 0.71	0.6816 ± 0.0011

Mean ± sample standard deviation across three seeds; not a confidence interval over future periods. Values are regenerated from saved predictions in the report build. [†]Added during this audit: always output $p = 807/1475 = 0.5471$, fitted only on training labels. Original trained runs are unchanged.

Interpretation. Reduced FININ is only 0.32 percentage points above always-up accuracy. Its balanced accuracy is approximately chance. A constant training-prior probability also nearly matches its log loss. This experiment provides no convincing evidence of an attention benefit on the substituted dataset.

Why accuracy alone is misleading

Full-model seed	Predict up	Predict down	Correct down	Balanced acc.
17	317	0	0 / 133	50.00%
42	317	0	0 / 133	50.00%
73	312	5	4 / 133	51.23%

There are 184 up days and 133 non-up days. Always predicting up therefore earns 58.04% accuracy while recovering none of the non-up days. Balanced accuracy averages recall across the two classes. Log loss assesses probabilities; assigning probability 1 to every day creates an extreme penalty on non-up days, so the train-prior reference is the useful probability comparison.

What each comparison tests

Mean pooling removes both attention mechanisms but retains the fusion pipeline. **No self-attention** retains market-query weighting. **No sentiment** replaces the three sentiment inputs with zeros while preserving the network shape. The two numeric baselines are simpler independent models; they are not exact replicas of the paper's ablations.

Connection to the publication. The paper reports its strongest Sharpe ratios at 20 input days: 1.772 on S&P 500 and 1.449 on NASDAQ 100 [1, Table 2]. These are contextual reference values, not benchmarks for the numbers above: market, data, timing, lookback and evaluation all differ. Our classification result does not confirm or refute that financial-performance claim.

04 / IMPLEMENTATION AND VERIFICATION

The code skeleton and evidence behind it

End-to-end workflow. Acquire pinned sources → audit dates and returns → build aligned examples → cache frozen features → train baselines/ablations → export predictions and presentation material.

Code area	Role and connection to FININ
<code>src/data/nifty.py, prices.py, examples.py</code>	Load verified snapshots; select headlines; align the trading calendar; derive labels; purge split boundaries and standardize inputs.
<code>src/features/cache.py</code>	Frozen BGE and sentiment extraction; cache provenance and hashes;
<code>src/training_data.py</code>	join arrays to dates; pad to 16 headlines and return masks.
<code>src/models/encoders.py</code>	Shared 384 → 64 text projection; distinct 6 → 64 market and 3 → 64 sentiment MLPs; separate 128 → 64 fusion MLPs. Paper Eqs. 1–5.
<code>src/models/attention.py</code>	One news self-attention layer, then a market query over refined news keys; masked softmax and weighted sum. Paper Section 4.2 / Eq. 6.
<code>src/models/finin.py</code> <code>src/models/baselines.py</code>	Concatenate market/news summaries; predict one logit; expose weights and structural variants. Reduced Eq. 7 with $t = 1$.
<code>src/training.py</code> <code>src/evaluation.py</code>	Seeded optimization, validation selection, checkpoint replay; probabilities, class recalls, confusion matrix and log loss.
<code>scripts/01_... through 06_...</code> <code>scripts/08_verify_prototype.py</code>	Ordered execution and offline export; new non-destructive audit reconstructs data and replays every checkpoint.

The central computation

$$\tilde{r}_i = \text{LayerNorm}(r_i + \text{SelfAttn}(R)_i), \quad \alpha_i = \text{softmax}_{i \in \text{real news}} \left(\frac{q(m)^\top k(\tilde{r}_i)}{\sqrt{64}} \right),$$

$$z = \sum_i \alpha_i \tilde{r}_i, \quad \hat{p} = \sigma(\text{MLP}([m; z])).$$

Implementation choices. The residual connection and LayerNorm are our concrete self-attention design choices; they are not established author defaults. Frozen encoders run outside training. Attention weights describe model allocation within a day and do not establish causal market influence.

Audit completed on 4 October 2026

Check	Measured outcome
Source and preparation integrity	2,111 source returns checked; all 2,107 saved examples match reconstruction; source/processed/cache hashes verified.
Feature and model validity	32,286 finite 384-vectors; valid three-way probabilities; seven integrity tests pass, including padding invariance and attention gradients.
Saved experiment replay	18/18 checkpoints reproduce all 317 test probabilities exactly on CPU; validation losses, selected epochs and saved attention diagnostics agree.
Optimization smoke check	Full model fits the first eight training examples: loss 0.6796 to below 0.00001 in 150 steps, with dropout disabled; not evidence of generalization.

Evidence: `reports/prototype_audit.json`, `reports/results.md`, saved run JSON/checkpoints. Environment: Python 3.12.0, PyTorch 2.5.1+cpu, NumPy 1.26.4. The trained architecture required no repair after this review.

05 / RESEARCH POSITION AND NEXT STEPS

What we can defend in the meeting

Claim supported now. An audited implementation preserves FININ’s fusion and attention mechanisms and runs a controlled experiment on substitute data, with baselines and ablations.

Claim not supported yet. Published gains, a forecasting benefit from attention, a trading strategy, and long-memory or market-inefficiency claims remain unverified.

Research observations that shape the next experiment

- **The paper’s primary success criterion is financial performance.** Its accuracy rankings differ from its Sharpe-ratio rankings. Thus a higher raw accuracy on our data would not itself reproduce the central claim [1, Sections 5.5–6.2].
- **Lookback length is central to the hypothesis.** FININ aggregates daily market/news representations over up to 20 trading days. Our model tests same-bucket headline interactions but omits the cross-day input history needed to study persistence [1, Eq. 7].
- **Data substitutions change the available signal.** Generated sentiment, capped coverage, a fixed market description and missing release times differ from TRNA. Current experiments do not isolate their individual effects.
- **Retrospective representation is a limitation.** BGE-v1.5 was released in 2023 [3], after our historical test period. Pretraining and sentiment-training provenance have not been audited for historical overlap. This is an offline historical experiment, not a point-in-time simulation.

Recommended next milestone

	Next action	What it would establish
1	Freeze the current version; diagnose class collapse using training/validation predictions and learning curves.	Separate optimization or calibration issues from weak usable signal; avoid tuning on the already-inspected test dates.
2	Add 3/5-day inputs, then 10/20 days if feasible; keep a matched mean-pooling baseline and register the comparison before evaluation.	Test whether temporal context changes the value of attention. Use fresh temporal holdouts or a declared rolling-window study.
3	Seek institutional TRNA access or timestamped, instrument-linked news; add a second market.	Move toward the paper’s input semantics and test whether findings persist across markets and periods.
4	Specify execution time, costs and risk-free-rate units before adding financial metrics.	Make PnL/Sharpe interpretable and reproducible. Reconcile Appendix B’s correctness-based flag and daily rate convention with an executable strategy.

Discussion point for the professor. Is the next milestone an architectural proof of concept on accessible data, or a closer empirical replication using licensed news and the original temporal evaluation? The current project is a verified foundation for the first, with a clear path toward the second.

References and provenance

[1] M. Wang, S. B. Cohen and T. Ma (2024). *Modeling News Interactions and Influence for Financial Market Prediction*. Findings of EMNLP, 3302–3314. doi:10.18653/v1/2024.findings-emnlp.189. Reviewed full paper, Figure 2, Eqs. 1–9, Tables 1–3 and Appendices A–C.

[2] R. Saqur. [NIFTY dataset card and files](#). Snapshot 9b8aef736cba; separate Yahoo SPY snapshot and checksums recorded in data/spy_manifest.json.

[3] BAAI. [BGE-small-en-v1.5 model card](#). Snapshot 5c38ec7c405e; 384-dimensional frozen first-token features.

[4] mikeysharma. [Finance sentiment analysis model card](#). Snapshot 4353ecd593de; ONNX TinyBERT; positive, neutral and negative outputs reordered and checked with control headlines.

Model/dataset cards consulted 4 October 2026. Full revisions and hashes are retained in project manifests. All prototype numbers are derived from local artifacts; no paper score is presented as our measurement.